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Section _____

MTH:2310 DISCRETE MATH
QUIZ 8
DUE: FRIDAY NOV. 14

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

Problem 1. (10 points) *Primes and Greatest Common Divisors*

- (a) (6 points) Find the greatest common divisor of 65 and 231 using the Euclidean algorithm. Show every step.

$$\begin{aligned} \gcd(231, 65) &= \gcd(65, 231 \bmod 65) = \gcd(65, 36) \\ &= \gcd(36, 65 \bmod 36) = \gcd(36, 29) = \gcd(29, 5) = \gcd(5, 4) = 1 \end{aligned}$$

(b) (4 points) Find the greatest common divisor and least common multiple of 65 and 231 using their prime factorizations.

$$65 = 5 \cdot 13$$

$$231 = 3 \cdot 7 \cdot 11$$

$\gcd = 1$ & there are no common factors

$$[M = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13]$$

Problem 2. (10 points) Solving Linear Congruences

Solve the following system of linear congruences using the construction method. Show every step.

$$\begin{cases} x \equiv 2 \pmod{3} \\ x \equiv 1 \pmod{4} \\ x \equiv 2 \pmod{7} \end{cases}$$

$$65 \pmod{3} = 2$$

$$65 \pmod{4} = 1$$

$$65 \pmod{7} = 2$$

$$N = 3 \cdot 4 \cdot 7 = 84$$

$$N_1 = \frac{84}{3} = 28 \quad 28M_1 \equiv 1 \pmod{3} \quad M_1 = 1$$

$$N_2 = \frac{84}{4} = 21 \quad 21M_2 \equiv 1 \pmod{4} \quad M_2 = 1$$

$$N_3 = \frac{84}{7} = 12 \quad 12M_3 \equiv 1 \pmod{7} \quad M_3 = 3$$

$$2 \cdot 28 \cdot 1 + 1 \cdot 21 \cdot 1 + 12 \cdot 2 \cdot 3 \pmod{84} = 56 + 21 + 72 \pmod{84} = 65$$

Problem 3. (10 points) Solving Linear Congruences

Solve the following system of linear congruences using the substitution method. Show every step.

$$\begin{cases} x \equiv 4 \pmod{5} \\ x \equiv 2 \pmod{7} \\ x \equiv 1 \pmod{9} \end{cases}$$

5, 7, 9 are pairwise prime, so we can use the construction method.

$$x = 4 + 5k \quad k \in \mathbb{Z}$$

$$x \equiv 2 \pmod{7} \Rightarrow 4 + 5k \equiv 2 \pmod{7} \Rightarrow 5k \equiv -2 \pmod{7} \Rightarrow 5k \equiv 5 \pmod{7}$$

$$k \equiv 1 \pmod{7} \Rightarrow k = 1 + 7g \quad g \in \mathbb{Z}$$

$$x = 4 + 5(1 + 7g) = 9 + 35g$$

$$x \equiv 1 \pmod{9} \Rightarrow 9 + 35g \equiv 1 \pmod{9} \Rightarrow 35g \equiv -8 \pmod{9} \Rightarrow 35g \equiv 1 \pmod{9} \Rightarrow 8g \equiv 1 \pmod{9}$$

$$g \equiv 8 \pmod{9} \Rightarrow g = 8 + 9m \quad m \in \mathbb{Z}$$

$$x = 9 + 35(8 + 9m) = 289 + 315m$$

$$x \equiv 289 \pmod{315}$$

Problem 4. (6 points) *Basics of Counting*

How many bit strings are there of length at most 5 that end with a 1 (not including the empty string)? Clearly show enough work to justify your answer.

$$\overbrace{3} \quad \overbrace{3} \quad \overbrace{3} \quad \overbrace{\cancel{0/1}/1} \quad \overbrace{1}$$

by the product rule,

there exist 3^4 bit strings of length at most 5 that end w/
2 1

Problem 5. (6 points) *Basics of Counting*

How many natural numbers are less than or equal to 360 but not a multiple of 9? Clearly show enough work to justify your answer.

there exist 360 natural numbers less than or equal to 360.

as $\frac{360}{9} = 40$, there exist 40 natural numbers less than or equal to 360 that are a multiple of 9.

thus the number of natural numbers less than or equal to 360 and not a multiple of 9 can be found using the subtraction rule.

$$360 - 40 = 320$$

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Problem 6. (6 points) Pigeonhole Principle

A roller coaster train can seat a total of 36 riders. The roller coaster was tracked for a span of two hours. In that time, the ride was operated 17 times and had 597 riders. Explain why the roller coaster must have been operated at least once at full capacity.

Suppose for contradiction no run operated at full capacity.

Then, by the pigeonhole principle, each run could have had at maximum 35 riders. $35 \cdot 17 = 595$. As $595 < 597$, the true number of riders, at least one ride must have been full.

